EMAIL/SMS SPAM DETECTION USING DL AND LLM

Submitted by

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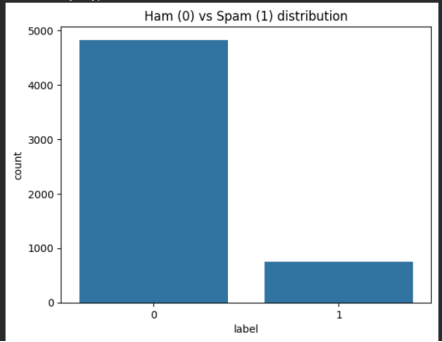
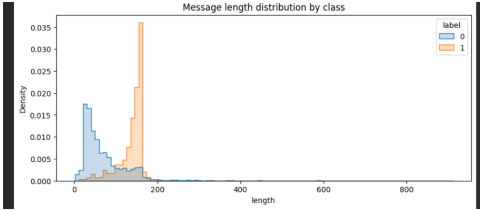
Problem Statement and Objectives

- **Problem Statement**
  - Rapid growth of SMS communication has increased unsolicited and malicious spam messages.
  - Existing filters struggle with evolving spam patterns, leading to security and privacy risks.
- **Objectives**
  - Design an accurate and robust SMS spam classification system.
  - Leverage deep learning to capture contextual and semantic patterns in text data.

Methodology / Approach

```
label
0    4825
1     747
Name: count, dtype: int64
```

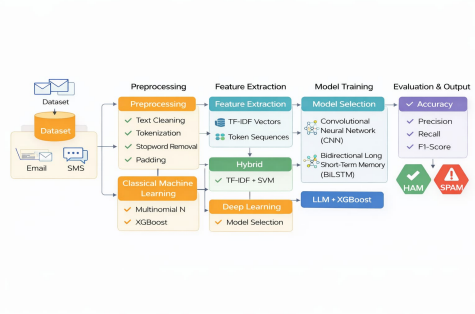




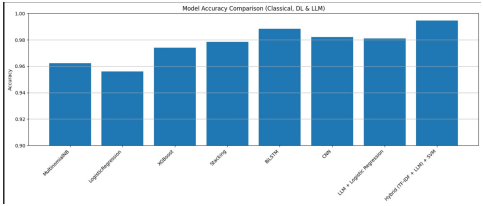
- SMS dataset preprocessing including cleaning, normalization, and tokenization.
- Feature representation through numerical encoding of textual data.
- Training and validation using advanced deep learning architectures.

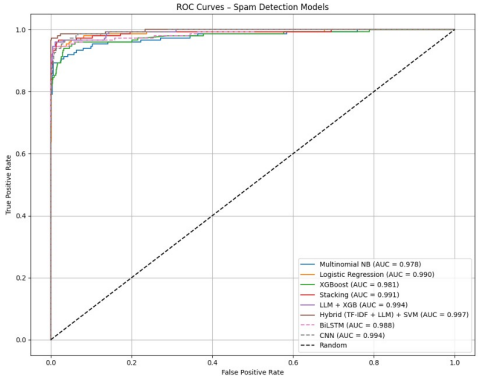
Code Working

- Raw Email/SMS text is cleaned, tokenized, and normalized.
- Text is transformed into TF–IDF vectors and token sequences.
- Classification is performed using:
  - Classical ML and hybrid models.
  - Deep learning models (CNN, BiLSTM).
  - LLM-assisted feature integration with XGBoost.
- The best-performing model outputs the final **Ham/Spam** prediction.



Achievements and Results

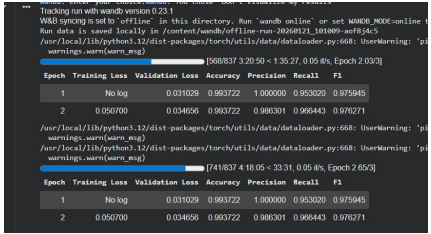




- Achieved overall accuracy of **0.994619**.
- Deep learning models outperformed traditional ML models.

Future Work

- **Multilingual Spam Detection**
  - Extend the current English-only system to multilingual text using language-agnostic models.
- **Online and Incremental Learning**
  - Enable continuous adaptation to evolving spam patterns without full model retraining.
- **Advanced Context-Aware Models**
  - Improve semantic understanding using domain-adapted transformer-based architectures.
- **Multi-Class Spam Classification**
  - Extend binary classification to identify specific spam such as scams.



Individual Contribution

- **Sambhav Ray**: Design and configuration of ML, deep learning, and LLM architecture along with model optimization and overall system integration.
- **Harsh Choudhary**: Performance evaluation, result analysis, interpretation of metrics, and technical documentation of findings.
- **Siddharth Dash**: Text preprocessing, pipeline development including cleaning, normalization, tokenization, and dataset preparation.
- **Harshvardhan Kintali**: Documentation and PPT preparation along with key metrices evaluation.
- **Tanvi Tulika Padhi**: Dataset understanding, exploratory data analysis support, data organization, and validation assistance.
- **Anshuman Samantray**: Data analysis support, dataset sorting, preprocessing verification, and conceptual understanding of workflow.
- **Ansuman Rout**: Dataset handling, data inspection, exploratory analysis assistance, and contribution to project understanding.

References

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